# SOFTWARE QUALITY ENGINEERING – LAB TASK 03

**Name : M. Haseeb Usman** **Reg No : SP24-BSE-137**

*Topic: CMMI Maturity Levels (Level 1 to Level 3) & Quality Assurance Implementation*

## 1. LAB ACTIVITIES

### Activity 1: Requirements Review Summary

- **Ambiguities:** Requirements use subjective terms like 'fast performance' or 'user-friendly' without clear, testable metrics.
- **Missing Requirements:** Critical scenarios like exception handling, offline mode, system timeouts, and edge cases are absent.
- **Inconsistencies:** Conflicting details across modules regarding business logic and data calculation steps.

### Activity 2: Student Grading System Scenario Analysis (Tasks 1–3)

- **1. Live System Edits:** Reflects chaotic practice. Updating code directly on production leads to unpredictable bugs. Remediation: Implement Configuration Management (CM) with version control and release baselines.
- **2. Undocumented Requirements:** Reflects non-standardized practice. Tacit knowledge leads to rework and scope mismatches. Remediation: Implement Requirements Management (REQM) with a documented Requirements Traceability Matrix (RTM).
- **3. Customer-Driven Testing:** Reflects reactive management. Fixing issues post-complaint is costly and bad for quality. Remediation: Implement PPQA & Project Planning (PP) to enforce pre-deployment test plans and execution.
- **4. Inconsistent Coding Styles:** Reflects lack of standardization. Makes code maintenance and onboarding difficult. Remediation: Introduce standard coding guidelines and pre-commit review checklists.
- **5. Informal Bug Tracking via Email:** Reflects chaotic practice. Issues get overlooked and defect trends remain unmeasured. Remediation: Implement Measurement & Analysis (MA) with a centralized issue tracking system (e.g., Jira).

## 2. HOMEWORK: MULTI-LEVEL SCENARIO ANALYSIS & TRANSFORMATION

### Task 1: Defect & Process Identification (EduTech Solutions)

- **Immediate Feature Implementation:** Chaotic/Reactive – No impact assessment or formal change control.
- **Gut Feeling Estimations:** Chaotic/Unpredictable – Lack of historical metrics leads to deadline slippage.

- **Non-Standard Folder Structures:** Lack of Standardization – High dependency on individual heroics.
- **Unvetted Subcontractors:** Chaotic/Lack of Control – No formal supplier quality verification.
- **Ad-hoc Pre-release Testing:** Reactive – Superficial testing right before release without test plans.

**Task 2: Level 2 (Managed) Remediation Plan Table**

| # | Identified Failure | Level 2 Process Area | Concrete QA Mechanism |
|---|---|---|---|
| 1 | Direct implementation of feature requests without checking existing code. | REQM (Requirements Management) | Establish a Requirement Change Control Form & Traceability Matrix (RTM). |
| 2 | Project estimation based on 'gut feeling' without recording metrics. | PP (Project Planning) | Implement a Project Estimation Worksheet based on WBS and historical data. |
| 3 | Inconsistent folder structures and naming conventions across developers. | CM (Configuration Management) | Create a Repository Setup Checklist & standard project folder template. |
| 4 | Hiring subcontractors on short notice without quality checks. | SAM (Supplier Agreement Management) | Establish a Supplier Evaluation Audit Checklist and acceptance review process. |
| 5 | Quick manual walk-through right before deployment. | PPQA / VER (Quality Assurance & Verification) | Enforce a Pre-Release Testing Sign-Off Checklist & Defect Tracking Log. |